

Week 5: SSH Key Authentication (Part 1)

Date: April 6, 2026

Study Time: 5-8 hours total

GOAL

Enable SSH key authentication between tier 1 and tier 2

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 10: Network Applications and Services (SSH section)

Video Learning

- ☐ LearnLinuxTV - Search: "SSH Key Authentication"
 - <https://youtube.com/@LearnLinuxTV>
- ☐ Professor Messer - Search: "SSH Keys"
 - <https://youtube.com/@professormesser>

Concepts to Understand

- ☐ Public key cryptography (asymmetric encryption)
 - ☐ Public vs private keys
 - ☐ How SSH key authentication works
 - ☐ `~/.ssh/authorized_keys` file
 - ☐ ssh-agent for managing keys
 - ☐ Key types: RSA, Ed25519, ECDSA
-

LAB WORK (3-4 hours)

Understanding SSH Keys

- ☐ Check existing SSH keys

bash

```
ls -la ~/.ssh/
# Common files:
# id_rsa / id_ed25519 = private key
# id_rsa.pub / id_ed25519.pub = public key
# authorized_keys = keys that can log into THIS machine
# known_hosts = fingerprints of servers you've connected to
```

Generate SSH Key Pair

☐ Generate Ed25519 key (recommended)

bash

```
ssh-keygen -t ed25519 -C "your_email@example.com"

# You'll be prompted:
# - File location (press Enter for default)
# - Passphrase (recommended for security)

# This creates:
# ~/.ssh/id_ed25519 (private - NEVER share!)
# ~/.ssh/id_ed25519.pub (public - safe to share)
```

☐ Alternative: Generate RSA key

bash

```
ssh-keygen -t rsa -b 4096 -C "your_email@example.com"
# -b 4096 = 4096 bit key size
```

☐ View your public key

bash

```
cat ~/.ssh/id_ed25519.pub
# Starts with "ssh-ed25519 ..."
# This is safe to copy anywhere
```

Copy Key to Server (Easy Way)

☐ Use ssh-copy-id

bash

```
# From your laptop to Fedora server:
ssh-copy-id user@your-fedora-server
# Enter your password one last time

# This automatically adds your public key to
# ~/.ssh/authorized_keys on the server
```

☐ Test passwordless login

bash

```
ssh user@your-fedora-server
# Should NOT ask for password!
# (May ask for key passphrase if you set one)
```

Copy Key to Server (Manual Way)

☐ If ssh-copy-id doesn't work

bash

```
# 1. Copy your public key to clipboard
cat ~/.ssh/id_ed25519.pub

# 2. SSH into server with password
ssh user@your-fedora-server

# 3. Create .ssh directory if it doesn't exist
mkdir -p ~/.ssh
chmod 700 ~/.ssh

# 4. Add your public key
nano ~/.ssh/authorized_keys
# Paste your public key
# Save and exit

# 5. Set proper permissions
chmod 600 ~/.ssh/authorized_keys

# 6. Exit and test
exit
ssh user@your-fedora-server # Should work without password
```

Set Up Keys for Tier 2 Devices

☐ Ubuntu laptop → Fedora server

bash

```
ssh-copy-id user@fedora-server-ip  
ssh user@fedora-server-ip # Test
```

☐ Windows desktop → Fedora server

bash

```
# If using WSL or Git Bash:  
ssh-keygen -t ed25519  
ssh-copy-id user@fedora-server-ip  
  
# Or use PuTTYgen and copy manually
```

☐ iPhone Termius → Fedora server

```
# In Termius app:  
# 1. Go to Keychain  
# 2. Generate new key  
# 3. Copy public key  
# 4. SSH to server, paste into authorized_keys
```

Verify Permissions

☐ Check file permissions (CRITICAL)

bash

```
# On the SERVER:  
ls -la ~/.ssh/  
  
# Correct permissions:  
# drwx----- (700) ~/.ssh/  
# -rw----- (600) ~/.ssh/authorized_keys  
# -rw----- (600) ~/.ssh/id_ed25519 (private key)  
# -rw-r--r-- (644) ~/.ssh/id_ed25519.pub (public key)  
  
# Fix if needed:  
chmod 700 ~/.ssh  
chmod 600 ~/.ssh/authorized_keys  
chmod 600 ~/.ssh/id_ed25519
```

Using ssh-agent

☐ Start ssh-agent

```
bash

eval "$(ssh-agent -s)"
# Starts agent and sets environment variables
```

☐ Add your key to agent

```
bash

ssh-add ~/.ssh/id_ed25519
# Enter passphrase once
# Now you won't be prompted again this session
```

☐ List loaded keys

```
bash

ssh-add -l
```

FLASHCARD COMMANDS

Create flashcards for these commands:

1. `ssh-keygen`
2. `ssh-copy-id`
3. `ssh`
4. `ssh-add`
5. `ssh-agent`
6. `cat ~/.ssh/authorized_keys`

Key Concepts:

- What's the difference between public and private keys?
- Where does the public key go?
- Where does the private key stay?
- What permissions should ~/.ssh have?

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 1.5 Given a scenario, use the appropriate networking tools
 - 3.2 Given a scenario, implement identity management
 - 3.4 Given a scenario, implement and execute security best practices
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NOTES & REFLECTIONS

SSH connections established:

- ☐ Laptop → Fedora server
- ☐ Desktop → Fedora server
- ☐ iPhone → Fedora server

Issues encountered:

Permission problems fixed:

Completed: ☐

QUICK REFERENCE

SSH Key Types (best to worst):

1. **Ed25519** - Modern, fast, secure (recommended)
2. **ECDSA** - Good, but some controversy
3. **RSA 4096** - Older but reliable
4. **RSA 2048** - Minimum acceptable

Important Files:

- `~/.ssh/id_ed25519` - Your PRIVATE key (never share!)
- `~/.ssh/id_ed25519.pub` - Your PUBLIC key (safe to share)
- `~/.ssh/authorized_keys` - Public keys that can log into this machine
- `~/.ssh/known_hosts` - Fingerprints of servers you trust
- `~/.ssh/config` - SSH client configuration (next week!)

Troubleshooting:

```
bash

# Verbose SSH output
ssh -v user@server
ssh -vv user@server    # More verbose
ssh -vvv user@server   # Maximum verbosity

# Check server SSH logs
sudo tail -f /var/log/auth.log # Ubuntu
sudo journalctl -u sshd -f     # Fedora
```

Security Tips:

-  Always use a passphrase on private keys
-  Keep private keys on your device only
-  Use Ed25519 for new keys
-  Never email or share private keys
-  Don't use keys without passphrases for important servers

Previous Week: sudo Configuration

Next Week: SSH Hardening & Deployment